

(TR-104) PYTHON WITH AI/ML

Training Day 85 Report

Date: 8 May 2026

Topic Covered: Backend Refactoring, Schema Synchronization, API Optimization & System Stability

The eighty-fifth day of training focused on backend refactoring, schema synchronization, bug fixing, API optimization, and improving overall system stability and user experience.

- Refactored the backend codebase using professional naming conventions and standardized coding practices.
- Improved synchronization between agent, router, tools, and formatter modules.
- Aligned backend services with the latest Customer, LeasePayment, and Insurance schema structures.
- Enhanced code readability, maintainability, and overall backend organization.
- Resolved runtime errors related to query processing and summary generation.
- Fixed Groq API tool parsing issues and improved handling of unexpected tool call formats.
- Implemented fallback mechanisms to prevent invalid or raw responses when datasets are empty.
- Improved backend response stability and error handling processes.
- Enhanced user experience by implementing dynamic fallback messages for empty search results.
- Added automatic UI behavior adjustments for scenarios where no data is available.
- Improved keyword recognition and support for lease-related queries.
- Analyzed payment-related database schemas and added support for the new “Partial” payment status.
- Updated payment processing, filtering, and query handling logic to support all payment states.

- Synchronized customer collection updates by integrating newly introduced schema fields.
- Updated formatter logic and tool responses to accurately reflect the latest customer schema.
- Performed frontend query testing to validate schema synchronization and response accuracy.
- Optimized API performance by reducing processing limits and improving payload management.
- Enhanced database fallback handling and improved reliability during large data processing operations.
- Revised concepts related to backend refactoring, schema synchronization, API optimization, fallback mechanisms, payment status management, and AI response stabilization.

Overall, the training enhanced understanding of enterprise-level backend maintenance, schema evolution, API reliability, system stability, and scalable AI-powered application development.